Infinite Series 2

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Problem 1

Assume $\sin(z) = 0$. Then we have

$$\sin(z) = \frac{e^{iz} - e^{-iz}}{2i} = 0$$

$$e^{iz} = e^{-iz}, \quad \implies \quad e^{2iz} = 1.$$

If $e^w = 1$, then $w \in 2\pi i\mathbb{Z}$. So we have that $2iz = 2\pi in$ for some $n \in \mathbb{Z}$, which gives us $z = \pi n$.

Problem 2

We can write $\zeta(k)$ as

$$\zeta(k) = \sum_{n=1}^{\infty} \frac{1}{n^k} = \sum_{\text{odd } n} \frac{1}{n^k} + \sum_{\text{even } n} \frac{1}{n^k} = \sum_{\text{odd } n} \frac{1}{n^k} + \frac{1}{2^k}\zeta(k).$$

So we can write the sum as

$$\sum_{n=1}^{\infty} \frac{1}{(2n-1)^k} = \sum_{\text{odd } n} \frac{1}{n^k} = \zeta(k) - \frac{1}{2^k}\zeta(k) = \zeta(k)(1 - 2^{-k}).$$

Similarly,

$$\zeta_a(k) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^k} = \sum_{\text{odd } n} \frac{1}{n^k} - \sum_{\text{even } n} \frac{1}{n^k},$$

$$= \sum_{n=1}^{\infty} \frac{1}{(2n-1)^k} - \frac{1}{n^k}\zeta(k) = (1 - 2^{-k})\zeta(k) - 2^{-k}\zeta(k) = \zeta(k)(1 - 2^{1-k}).$$

So

$$\sum_{n=1}^{\infty} \frac{1}{(2n-1)^k} = \frac{1}{1 - 2^{1-k}}\zeta_a(k).$$

Problem 5

Part (a)

First, we compute the inner integral by setting $u = xy$, so $du = y\,xy$ and $dx = du/y$.

$$I = \int_0^1 \frac{dx}{1+(xy)^2} = \frac{1}{y}\int_0^y \frac{du}{1+u^2} = \frac{\arctan y}{y}.$$

Then, we substitute the power series of $\arctan u$:

$$\frac{\arctan y}{y} = \sum_{n=0}^{\infty} \frac{(-1)^n y^{2n}}{2n+1}.$$

Interchanging the sum and integral, we get

$$I = \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1}\int_0^1 y^{2n}dy = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)^2} = G.$$

Problem 9

Part (a)

$$\frac{1}{m^n} + \frac{1}{m^{2n}} + \frac{1}{m^{3n}}\cdots = \frac{1/m^n}{1-1/m^n} = \frac{1}{m^n-1}.$$

Part (c)

Swap the order:

$$\sum_{k=2}^{\infty}\sum_{n=2}^{\infty}\frac{1}{n^k} = \sum_{n=2}^{\infty}\sum_{k=2}^{\infty}\frac{1}{n^k}.$$

Now we evaluate the inner sum:

$$\sum_{k=2}^{\infty}\frac{1}{n^k} = \frac{1/n^2}{1-1/n} = \frac{1}{n(n-1)}.$$

So

$$\sum_{n=2}^{\infty}\sum_{k=2}^{\infty}\frac{1}{n^k} = \sum_{n=2}^{\infty}\frac{1}{n(n-1)} = \sum_{n=2}^{\infty}\left(\frac{1}{n-1}-\frac{1}{n}\right) = 1.$$

Problem 10

The denominator $m^n - 1$ is odd exactly when m is even. So we can evaluate the same sum from problem 9 but restrict m to even values:

$$\sum_{n \text{ even} \geq 2}\sum_{k=2}^{\infty}\frac{1}{n^k} = \sum_{n \text{ even} \geq 2}\frac{1}{n(n-1)} = \sum_{m=1}^{\infty}\frac{1}{(2m)(2m-1)}$$

$$= \sum_{m=1}^{\infty} \left(\frac{1}{2m-1} - \frac{1}{2m} \right) = 1 - \frac{1}{2} + \frac{1}{3} - \cdots = \ln 2.$$

Problem 13

By Theorem 2.1,

$$\pi \cot \pi x = \frac{1}{x} + \sum_{n=1}^{\infty} \frac{2x}{x^2 - n^2}$$

for all $x \in \mathbb{C} \setminus \mathbb{Z}$. Plugging in $x = i$, we get

$$\pi \cot(i\pi) = -i - 2i \sum_{n=0}^{\infty} \frac{1}{n^2+1},$$

$$\sum_{n=0}^{\infty} \frac{1}{n^2+1} = -\frac{\pi \cot(i\pi) + i}{2i} = -\frac{-i\pi \coth \pi + i}{2i} = \frac{\pi \coth \pi - 1}{2}.$$